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'*Los von London*': A comparative, empirical analysis of German and British global foreign banking and trade development, 1881–1913

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Abstract

The role of finance in the development of trade draws increasing attention from economists and economic historians. Yet empirical studies, especially from an historical perspective, continue to be scarce. This study analyses the role of German and British foreign banks in the internationalisation of trade during the first globalisation. It creates a novel data set on the bilateral trade of Germany and Great Britain with the rest of the world and the number and geographical distribution of German and British foreign banks between 1881 and 1913. Using an augmented gravity model of trade, the article shows that banks had a significant positive impact on exports and imports and that this effect was even more pronounced in case of German banks and trade. Moreover, the effect of German banks on trade is the highest in the years closer to bank entry, supporting the idea of German banks being initiators of trade. In contrast, the effect of British banks seems constant over time.

KEYWORDS

banks, finance, first globalisation, Germany, Great Britain, international trade

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After political unification in 1871, Germany entered a period of intense and rapid industrialisation, and its economy began to expand beyond national frontiers. Effectively, it became necessary for German commerce and industry to overcome the limitations of the supply of commodities, raw materials, and capital goods afforded by the home market.[1] The major global player in foreign trade at that time was Great Britain. Yet, since the 1880s Germany had been expanding its foreign commercial relations constantly, challenging British hegemony intensively. In the period between 1881 and 1913, the compound growth rate of German trade was 4.4 per cent, whereas British exports only grew by 2.4 per cent (see online appendix). Indeed, by the turn of the century, Germany had become Great Britain's most important competitor in world trade markets.[2]

This study examines the role of German and British banks in the development of German and British trade in the late nineteenth and early twentieth centuries. It argues that the rapid and successful expansion of German trade in international markets from the 1870s was facilitated by the establishment of the first *Auslandsbanken* (foreign banks), able to meet the informational and financial needs of the foreign commerce sector. The particularity of German foreign banking seems to have been the concentration of activities primarily in those markets where German trade was only marginally developed. This contrasts with the banking dynamics of economies that, by that time, already had established themselves in international commercial markets, such as that of Great Britain.[3] The motivation behind British banks going abroad in the 1860s and 1870s was based on intensification of investment flows and trade in overseas settlements.[4] Studies highlight the differences in information and credit policies, showing that German banks provided the export sector with more detailed information about international market conditions and seemingly granted longer periods of credit.[5] Moreover, studies on the German banks' entry into Latin America show that the *Auslandsbanken* prioritised their credit extension to export houses of German origin and/or companies exporting to Germany.[6]

The positive relationship between the access to finance and trade development has been shown by qualitative and empirical studies alike. Kletzer and Bardhan, for example, highlight the link between the quality of a country's financial system and the development of sectors and industries that depend on external finance, that is, the export sector.[7] They show that imperfections in credit markets, such as moral hazard and credit contract enforcement under incomplete information, may define a country's comparative advantage in international trade markets.[8] The perspective has been widened by Beck, showing that trade flows are positively correlated with the size of

[1] Hurley, *Banking and credit*, p. 18; Forbes, 'German informal imperialism', pp. 384–5.

[2] As stated by Neuburger and Stokes, 'The Anglo-German trade rivalry', p. 191: 'In 1897 the largest exporting country was the United Kingdom (US $1431.9 million) followed closely by the United States (US $1153 million), Germany (US $865.13 million), and then France (US $694.41 million). [...] The dominant position of the United Kingdom in exports at the beginning of the period was seriously eroded by the end. Although the United Kingdom still exported more than other countries, it appears that if the prevailing pattern continued, Germany would have overtaken the United Kingdom before long'.

[3] See for example Hauser, 'Die Deutschen Überseebanken'; Whale, *Joint stock banking*.

[4] Battilossi, 'Multinational banking', p. 7.

[5] Hurley, *Banking and credit*; Riesser, *German great banks*; Young, 'British overseas banking'.

[6] Kisling, 'Trade finance'.

[7] Kletzer and Bardhan, 'Credit markets'.

[8] See also Contessi and de Nicola, 'Finance and international trade', p. 7.

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a country's banking system,[9] while Bordo and Rousseau confirm that countries with a higher level of financial development have higher export shares and trade balances.[10] Studying the link between financial development and trade in the late 1990s and early 2000s, Turner affirms that the impact of finance on exports is defined by the reliance of the sector not only on external finance but also on the availability and efficiency of financial intermediaries. He finds a significant positive correlation between geographic bank penetration, in terms of total number of branches and export volumes.[11] Using firm-level data, Kisling shows that foreign bank entry in nineteenth-century Brazil eased previous existing credit constraints for export companies, and export companies financed by banks exported significantly more than companies not financed by banks.[12] Studies on nineteenth-century German and British international banking and trade developments contemplate different aspects. Some deal with wider perspectives, illustrating German and British international trade developments,[13] the emerging trade rivalry between the two nations,[14] and the general characteristics of German and British banking.[15] Other work analyses specific details: the influence of German and British foreign banks on the development of local banks,[16] the role of British banks in fostering investments in emerging markets,[17] the possible effects of competition between German and British banks,[18] and the individual history of selected banks.[19] Empirical analysis of the relation between global banking and trade dynamics comparing Germany and Great Britain is left wanting.

This study provides new insights and empirical evidence on the link between foreign banking and trade development in the cases of Germany and Great Britain between 1881 and 1913. It creates a new panel data base on (i) the bilateral trade of both Germany and Great Britain with the rest of the world, and (ii) the number and geographical distribution of German and British foreign banks abroad for the benchmark years 1881, 1900, and 1913. This article provides the first quantitative evidence of a direct link between foreign banking and trade during the first globalization. It tests the correlation between movements in exports and imports and banking penetration by running an augmented gravity model of trade. The augmented gravity model of trade, furthermore, accounts for possible differences in the magnitude of the relation between bank penetration and

[9] Bordo and Rousseau, 'Historical evidence', pp. 1236–7. Trade is measured as the sum of total exports and imports, and the size of the banking system is defined as the ratio of broad money to GDP.

[10] Beck, 'Financial development', p. 4. The indicators of trade in manufactured goods was defined as exports and imports and their difference relative to GDP. Financial development was measured as credit to the private sector by deposit money banks and other financial intermediaries as share of GDP.

[11] Turner, 'Financial geography', pp. 269–70, 279–80.

[12] Kisling, 'Trade finance'.

[13] Platt, *Finance, trade and politics*; Forbes, 'German informal imperialism'; Bulmer-Thomas, *Economic history of Latin America*.

[14] The Standard, *How to capture German trade*; Hoffmann, *Trade rivalry*; Neuburger and Stokes, 'The Anglo-German trade rivalry'.

[15] Riesser, *German great banks*; Born, *Geld und Banken*; Pohl, *Das deutsche Bankenwesen*; Tilly, 'Large German banks up to 1914'.

[16] Briones and Villela, 'European bank penetration'.

[17] Arndt, *Deutschlands Stellung in der Weltwirtschaft*; Halsey, *Investments in Latin America*; Rippy, 'German investments in Latin America'; idem, *British investments in Latin America*; Corley, 'Britain's overseas investments'.

[18] Young, 'British overseas banking'.

[19] Joslin, *A century of banking*; Pohl, *Deutsche Bank Buenos Aires*.

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trade growth over time, measuring the overall effects of bank presence and the specific effect of banks on exports and imports in each year.

The empirical results confirm a positive and significant correlation between the number of German and British foreign banks in a country and the respective trade with that country. The relationship appears stronger in case of Germany. Moreover, the impact of the German banks' presence seems to be more significant in 1881 than in 1913. The effect of British banks, in contrast, seems to be constant over time. These findings provide empirical evidence supporting the idea of contemporaneous qualitative studies that German banks acted as initial trade promoters. German banks were more likely to establish themselves in markets that showed a high demand for foreign goods but where German trade only had a marginal market share previous to bank entry.

The article is structured as follows. Section I discusses the dynamics and strategies of German banking internationalisation and trade finance in the late nineteenth and early twentieth centuries. Section II presents the data. Section III provides a comparison of German and British foreign trade and foreign banking development between 1880 and 1913. The specifications and the results of the econometric analysis, the augmented gravity model of trade, are presented in section IV. Section V concludes.

I | THE GERMAN STRATEGY OF TRADE FINANCE – '*LOS VON LONDON*'

> 'The object of the company is the transaction of all sorts of banking business, particularly the fostering and facilitating of commercial relations between Germany, the other European countries, and overseas markets.'
>
> (Denkschrift Deutsche Bank, 1869).[20]

The principal strategy of German banks in fostering the advancement of German trade in the international markets in the late nineteenth and early twentieth centuries was simple and ambitious at the same time: '*Los von London*' (away from London), or become financially independent from Great Britain. At the time, London was the 'clearing house' of international payments and the centre of trade finance. German industrial and commercial interests in foreign countries had to rely nearly entirely on the intermediation of English banks.[21]

The financial supremacy of Great Britain derived most importantly from two facts. First, with sterling, Great Britain possessed the most stable and internationally accepted currency in the world. Second, the pillars of the British financial system were the old banking houses with established reputations, rich experience, and a high level of clients' trust.[22] Moreover, they were able to

[20] Translated from German by author. See also Riesser, *German great banks*, p. 421; Hauser, 'Die Deutsche Überseebanken', pp. 18–9; and Whale, *Joint stock banking*, p. 16.

[21] For two of many examples, see Hauser, 'Die Deutsche Überseebanken', p. 18; Frankfurter Zeitung, *Gegen die englische Finanzvormacht*.

[22] According to Einzig, *Financial supremacy*, pp. 26ff, the dominance of London can additionally be explained by, firstly, the existence of a well-developed money market that included a distinctive call money market – the sterling bill was the principal payment in international trade – a market for short-term loans, and a discount market unrivalled by any other market at this time. Secondly, the British financial market was highly liquid. In this period Great Britain possessed immense resources in the form of external investments, representing the accumulation of favourable trade balances. In

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narrow their profit margins considerably in comparison with banks of other nations, as London showed a larger banking turnover than any other financial centre at this time.[23] The advantage of British banks in narrowing their yields significantly is considered to be one of the major obstacles that their German counterparts, and with them foreign trade, faced. One of the consequences was that British discount rates were constantly beneath those offered in Berlin or Hamburg. London was simply the cheaper alternative for German merchants.[24]

Consequently, given the dominance of financial institutions and markets in Great Britain, German banks focussed on the creation of their own independent network of financial institutions in international markets: the *Auslandsbanken*.[25] In their aim to support German trade, they applied two approaches: first, providing German exporters and importers with adequate credit, both in foreign markets and in the home market, and second, fostering the establishment of the German currency in the international market.[26] One of the first *Auslandsbanken* was established in London in 1870, a subsidiary of Deutsche Bank. In the same year Commerz Bank, together with Disconto Gesellschaft, founded the London-based London and Hanseatic Bank. Soon other European countries, such as The Netherlands (Amsterdamsche Bank) in 1871 or Italy in 1894

addition, British government bonds and other securities, considered to be particularly safe, enjoyed high prestige among foreign investors. Thirdly, the high degree of liberalisation of the British financial market, being reflected by (i) the easy access of foreign banking to London with no discriminating legal conditions, only with the exception of being part of the clearing house, and no special establishment requirements, (ii) no legal restrictions regarding employment or (iii) in dealings in exchanges and international transfer of funds, and (iv) the absolute freedom of the gold market, with the Bank of England following a strict neutral policy with respect to gold movements. The latter was partly due to the high reserves and constant gold inflows from the British colonies in South Africa. Fourthly, there existed a large, wealthy class of investors willing to invest in foreign markets. This might be derived from their historical experience in foreign investments, with British colonies ever since having been the destination of British investments, and their trust in British financial institutions, which inhered the image of following rather risk-adverse investment policies.

[23] Ibid.

[24] In 1904, the bank discount rate offered in Berlin was, in 21 out of 24 cases, one per cent higher than the London rates; see Hauser, 'Die Deutsche Überseebanken', pp. 14–5. Moreover, contemporaneous calculations suggest that the German banks' loss of fees and commissions to their British competitors for handling German payment transactions accounted, until the late 1880s, for up to 3.5 million marks per year; see ibid., p. 17; Strasser, *Die Deutschen Banken im Ausland*, p. 23.

[25] German banks followed three main strategies to establish their presence in overseas markets: (i) the extension of their organisation to foreign countries, (ii) the construction of close relationships with foreign financial institutions, and (iii) the foundation of associated foreign banks as subsidiaries: the *Auslandsbanken.* See Whale, *Joint stock banking*, pp. 68–89; Strasser, *Die Deutschen Banken im Ausland*, pp. 37ff. The earliest technique was to enter into a 'sleeping' partnership (*Kommandite*) with a foreign institution, with the primary aim of strengthening economic ties with countries that already disposed over a sound domestic financial system. The leading *Grossbanken* was the first to implement such relationships. In 1870, the Darmstädter Bank formed a silent collaboration with the Vienna-based Dutschko & Co. See Riesser, *German great banks*, pp. 439, 987. Already, five years earlier, Deutsche Bank had established relations with the United States in the form of a commandite with the New York bank Knoblauch & Lichtenstein; see ibid., p. 987. Additionally, in some cases German and foreign banks established a partnership by agreeing upon cooperation on certain matters of interest but with full freedom of action. Similar to the *Kommandite*, this 'community of interest' was formed to enter markets with little necessity and/or scope for new institutions. Despite forging closer ties than those established by previous correspondent networks, the *Kommandite* and the community of interest ran the risk of the management remaining in foreign hands, which were unlikely to define German interests as their priority and, furthermore, tended to hazard German capital on their own speculations. Besides, such connections were limited to being formed with partners of relatively minor importance. In regions showing less development in financial terms and in maintaining commercial relations with Germany, German banks aspired to obtain more direct and distinctive control and created subsidiary banks. See Whale, *Joint stock banking*, pp. 66–9.

[26] See for example Riesser, *German great banks*, pp. 420–1; Hurley, *Banking and credit*, pp. 7, 19; Whale, *Joint stock banking*, pp. 66–7; Tilly, 'Germany', p. 106.

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(Banco Commerciale Italiana), followed.[27] After focussing initially on the establishment of German banks in the core European countries, the next step was to expand German presence to the emerging periphery, primarily countries in South America and Asia. The first overseas bank with German participation was established in Argentina. In 1872 the Disconto-Gesellschaft together with Belgian banking institutions founded the La Plata Bank. The bank was partially taken over by Deutsche Bank in 1874. Although they succeeded in expanding their operation to Uruguay, the life span of the bank was short. Extensive investments in Uruguayan government bonds in the 1870s led to an even more substantial loss after a regime change in 1874, when the new government in Uruguay refused to honour its debts. The situation was worsened by an intensive depreciation of silver in the late 1870s, diminishing invested capital. Never really recovering in the following years, Deutsche Bank finally managed, with great losses, to liquidate the bank in 1884–5.[28]

Soon after this first tentative and unfortunate attempt, in the early 1880s, a government-run *Überseeische Reichsbank* (1881–1884) was launched with the idea of establishing a single bank responsible for all German overseas banking activities. In addition to the German government's idea of gaining influence in the commercial ventures of German business men and banks, it clearly followed the aim to establish some sort of political influence in foreign countries via the *Auslandsbanken*. However, this idea was met with little enthusiasm amongst private bankers, who in fact hindered the initiative of an *Überseeische Reichsbank* in 1886.[29] Instead, the largest and most important institution, Deutsche Bank, founded in the same year, became one of the most influential *Auslandsbanken*, the Deutsche Überseeische Bank.[30] In general, the influence and involvement of German politics in foreign banking can be described as rather marginal. The expansion of German banks in foreign countries indeed was driven primarily by private and economic reasons.[31] Literally all the *Auslandsbanken* were founded by one of the *Grossbanken* (large banks)[32]

[27] Riesser, *German great banks*, p. 433–6, 985–7; Forbes, 'German informal imperialism', pp. 392–3.

[28] Jones, *Banks as multinationals*, p. 103; Riesser, *German great banks*, pp. 422, 433, 443; Young, 'British overseas banking', p. 89.

[29] Deutsche Ökonomist; Young, 'British overseas banking', p. 90.

[30] Deutsche Überseeische Bank, *Geschäftsberichte* (1886).

[31] A minor exception and an example of the intent of the German government to maintain some counterbalance to the private *Grossbanken* was the *Seehandlungsgesellschaft* (Maritime Trade Association), later called the Prussian State Bank. Founded in 1772 by Frederick II as a joint-stock company with the primary purpose of fostering commerce, with special privileges to trade transmarine salt and to finance state expenditures, it passed under the control of the Prussian Ministry of Finance in 1848. See Wirth, *Banking in Germany*, pp. 10–12. The most famous example may be its involvement in the foundation of the Deutsch-Asiatische Bank. Internal struggles among the leading *Grossbanken*, mainly the growing competition among the two largest institutions – the Deutsche Bank and the Diskonto-Gesellschaft – impeded the foundation of the China Bank created by a Bank Konsortium. It was the *Seehandlungsgesellschaft* that finally intervened and subscribed the capital of the new China Bank. Founded in 1889 it was named the Deutsch-Asiatische Bank. Besides the involvement of the *Seehandlungsgesellschaft*, it was also the Deutsche Bank, the Darmstädter Bank, the Diskonto-Gesellschaft, the Berliner Handelsgesellschaft, the A Schaafhausen'scher Bankverein, the Nationalbank für Deutschland, and the Dresdner Bank that participated in its foundation. See Strasser, *Die Deutschen Bank im Ausland*, pp. 29–30; Stiftung Seehandlung, 'Die Stiftung Preußische Seehandlung'; Backhaus, 'Banking institutions', p. 212; Hertner, 'German banks abroad', pp. 107–9.

[32] The term *Grossbanken* has been subject to several interpretations. The Deutsche Bank and the Dresdner Bank, together with the Disconto-Gesellschaft and Darmstädter Bank [*für Handel und Industrie* (for trade and business)] created in 1851 and 1853 are often referred to as the "*Vier D-Banken*" (four D-bank). Commonly, the concept *Grossbanken* describes the nine largest (in terms of capital and branches) German banks of the second half of the nineteenth and early twentieth centuries: A Schaafhausen'scher Bankenverein (1848), Disconto-Gesellschaft (1851), Darmstädter Bank (1853), Berliner Handelsgesellschaft (1856), Mitteldeutsche Creditbank (1856), Deutsche Bank (1870), Commerz Bank (1870), Dresdner

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or in cooperation between them in the form of a *Konsortium*. The latter may confirm the idea of historical scholars that one special feature of German banks was their ability to finance risky and capital-intensive ventures by cooperation.[33] This dynamic apparently was not only limited to the operations within Germany but also applied to German banking activities abroad. In 1886 the Deutsche Überseeische Bank was founded by Deutsche Bank, focussing on the Latin American market. Only one year later, the Brasilianische Bank für Deutschland followed, founded as a co-venture between Norddeutsche Bank and Diskonto-Gesellschaft. Subsequently, German banking expanded to the Asian continent. In the mid-1880s German industry, and with it the German *Grossbanken*, focused more and more on Asian markets. The German railway companies aimed to establish new business in China. In 1889, the previously discussed Deutsch-Asiatische Bank was created. In the early twentieth century Germany started to expand its presence in Africa. In 1905 the Deutsche Afrika Bank was founded, concentrating its activities on the Empire's colonies, such as Namibia and Tanganyika (in modern-day Tanzania) (German East Africa). In 1905, the Deutsche Orientbank followed, operating mainly in the Far East, competing directly with the British in Egypt, especially in becoming one of the principal financiers of the cotton sector.[34] Online appendix table A1 lists the German *Auslandsbanken* founded between 1869 and 1913, and their mother institutions.

The literature suggests that German banks failed in breaking the dominance of sterling or London bills as the key currency and primary access to cheap credit. Nevertheless, they were successful in converting the German mark into an international currency. Not being able to challenge the British pound, it became a crucial currency and possible alternative that enjoyed increasing utilisation in some European countries, such as Sweden, Italy, and Austria-Hungary, and in Asia and South America, which were the focus of German foreign banking activities.[35] However, the credit policies of the German banks and their *Auslandsbanken* are commonly defined in literature as the preeminent feature of their successful financing of foreign trade. The *Auslandsbanken* were known to grant comparatively longer credits than their competing counterparts from Great Britain or any other nation at that time. German banks discounted bills of exchange up to 240 days on credit,[36] and it was common practice that German banks paid German exporters up to 70 per cent of the total value of their delivery before the actual arrival of the same. They moreover granted *Kreditbriefe* (direct letters of credit) that permitted the exporter to receive payment at the moment the shipping and paying documents (*Konossement*) arrived, but before the actual arrival of the goods. The *Kreditbrief* was emitted by the *Auslandsbank* and sent to its mother institution based in Germany. At the time the *Konossement* arrived at the *Auslandsbank* in the country of destination of the exports, the German exporter was paid the full amount by the German mother institution.[37] Most interestingly, German banks seemed to act as some sort of 'buyer of last resort' for German exporters. In case of non-payment by the overseas importer, German banks granted an '*Akkzeptkredit*' (reimbursement credit) to the German exporter, with an amount up to 74 per cent of the invoice. Furthermore, they received and stored the shipment and administrated

Bank (1872), and the Nationalbank für Deutschland (1881) (see e.g. Riesser, *German great banks*, and Whale, *Joint stock banking*.).

[33] Schwanitz, *Gold, Bankiers und Diplomaten*; Riesser, *German great banks*, pp. 443, 446.

[34] Foxwell, 'The financing of industry and trade'.

[35] Tilly, 'Large German banks up to 1914', pp. 108–9.

[36] Strasser, *Die Deutschen Bank im Ausland*, p. 22; Whale, *Joint stock banking*, p. 87; Young, 'British overseas banking', p. 86.

[37] Hauser, 'Die Deutschen Überseebanken', p. 5.

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further sales.[38] Some studies even argue that the involvement of German banks in supporting the establishment of industrial companies abroad exceeded pure financing, as they provided the German home market with information on foreign trade and investment opportunities and eventually even took initiatives to found foreign business ventures themselves.[39] They thus served as some sort of 'pioneer' for German industry in foreign markets. They aimed to win new trade markets, in contrast to the objectives of banks from other nations, such as Great Britain, to accommodate already existing trade.[40] This argument is most interesting, as it creates a certain kind of causality between the development of German foreign trade and banking. At the same time, it somehow contradicts the literature on banking internationalisation, assuming that banks tend to follow their clients, in a chronological sense, and not vice versa.[41]

II | THE DATA

This paper introduces two new data sets. First, it creates a new panel dataset on German and British global bilateral trade for the periods from 1880–4, 1898–1902, and 1909–13 is created, on the basis of Federico and Tena-Junguito's existing dataset on British foreign trade[42] and drawn from hand-collected data on German foreign trade, taken from the official trade statistics of the German Empire, *Statistik des Deutsche Reiches – Auswärtiger Handel des deutschen Zollgebietes nach Herkunfts- und Bestimmungsländern.*[43] The number of German and British trade partners in the resulting data set varies over time. According to the official German trade statistics in 1881, Germany traded with 35 different countries – in 1900 with 71 and in 1913 with 79 countries, respectively. For British trade, Ferderico and Tena-Junguito report 54 different trade partners in 1881, 69 in 1900, and 72 in 1913.

The second dataset contains information on the universe of German and British foreign banks and their branches in the world on a city level for the years 1881, 1900, and 1913. The data were hand-collected from *The banking almanac, directory*, *year book and diary*.[44] The almanac was published annually from 1845 until 1919 (succeeded by the *Banker's almanac and year book*) and provides detailed information on the number and location of branches on a city level of all foreign banks in the world that had representation (branch, agent, or correspondent) in London. Further, the almanac provides selected balance sheet information, such as the amount of cash or bank deposits. However, this information is not provided consistently across banks and over time. To measure and account for the size of banks, I use the amount of their paid-in capital, since this information is uniformly provided for all years and for the largest sample of banks. The final data set includes the information of capital paid in for 55 per cent of all banks. In some cases, I could locate the original banking reports (*Geschäftsberichte*) of the *Auslandsbanken*, such as Deutsche Überseeische Bank and Deutsche Asiatische Bank. Furthermore, I verified the data from the

[38] Ibid., p. 6; Riesser, *German great banks*, pp. 246–8.

[39] Strasser, *Die Deutschen Bank im Ausland*, pp. 27–8; Tilly, 'Large German banks up to 1914'.

[40] Hurley, *Banking and credit*, p. 18.

[41] Buch, 'Why do banks go abroad?'.

[42] Federico and Tena-Junguito, *World trade, 1800–1938*. I thank Antonio Tena-Junguito for generously sharing the data before publication.

[43] Statistik des Deutsche Reiches, *Auswärtiger Handel.*

[44] The banking almanac.

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almanac and, if necessary, supplemented it with the information provided by the quantitative works of Hauser, Riesser, Hurley, Strasser, Lough, and Whale.[45]

We rely on bank data on three benchmark years, 1881, 1900, and 1913, for two reasons. Firstly, as the information is hand-collected, data collection is work-intensive. For the three years under study, the set includes 4896 raw observations of German and British bank branches around the world, at the city level. Secondly, the chosen benchmark years ensure the representativeness of the sample. With only a few exceptions, German foreign and overseas banks were founded after 1881. It was not before 1886 that the first German overseas bank was successfully established (see previous section). The year 1913 marks the dawn of the First World War and with it the end of the first period of internationalisation of modern banking and trade. The year 1900 has been chosen to account for possible changes in banking development and to capture the impact of banks on trade in the medium term.

The third data set combines the information on German and British global foreign banking and trade for the years 1881, 1900, and 1913. The choice of years mirrors the available bank data. The data set includes aggregated information on the number of unique banks and their branches per country, and the total value of German and British exports to and imports from a country. Moreover, it provides the GDP of the two countries and their trade partners and the geographical distance between them (see appendix). To address the difference in the reported countries in the German and British original trade statistics over time, a balanced data set was created that includes 104 different countries at any given year for both Germany and Great Britain. A country is included when it traded with and/or had a bank relationship with Germany and/or Great Britain in at least one of the three years: 1881, 1900, or 1913. Countries that appear in the balanced data set but not in the official German and/or British trade statistics in a certain year are assumed to have had zero trade with Germany and/or Great Britain in that year; the same is true for banking presence. However, the available information on GDP as provided by the Maddison Project Database[46] is limited. The resulting final data set used for the econometric analysis includes 334 total observations, consisting of 53 countries in 1881 and 57 countries in 1900 and 1913, respectively.

III | GERMAN AND BRITISH FOREIGN BANKS AND WORLD TRADE

The aim of this section is to empirically analyse and compare German and British foreign trade and banking development in the late nineteenth and early twentieth century. Figure 1 shows the development of German and British exports between 1880 and 1913.

The British were the dominant force in international trade markets when Germany started to intensify its foreign trade in the 1880s, and they continued to dominate during the entire period of investigation. In 1880, German foreign commerce accounted for a total value of 1.104 million USD. British trade, with 3.366 million USD, was valued at more than three times as much (see online appendix tables A2 and A5). However, having reached a relatively high level of maturity, British trade grew far less impressively than that of Germany. Between 1880 and 1913 German trade grew at a rate of 4.33 per cent, and British trade merely at 2.35 per cent. Especially in the period between 1880 and 1884, Germany managed to catch up, showing an annual growth rate above 3.68 per cent

[45] Hauser, 'Die Deutschen Überseebanken'; Riesser, *German great banks*; Hurley, *Banking and credit*; Strasser, *Die Deutschen Banken im Ausland*; Lough, *Banking opportunities*; and Whale, *Joint stock banking*.

[46] Bolt and van Zanden, *The first update*.

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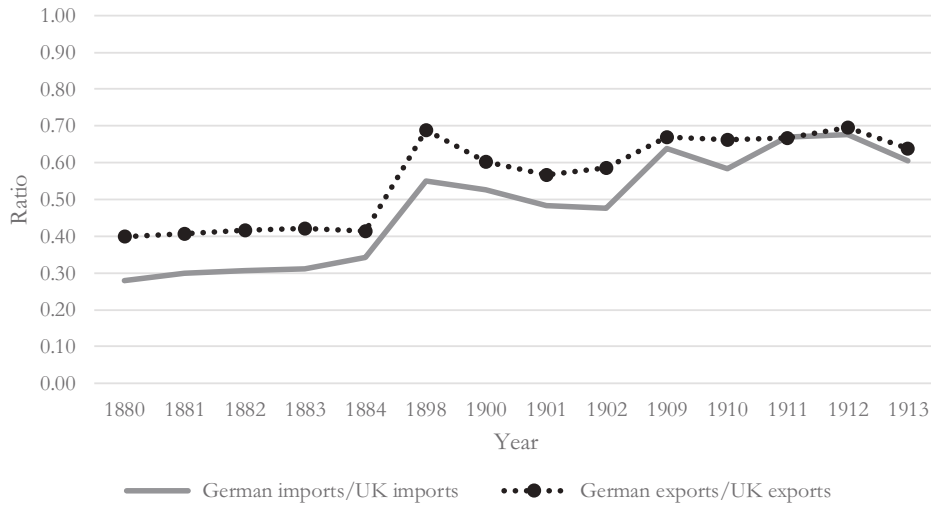

FIGURE 1 German exports and imports relative to British exports and imports – 1880–4, 1898–1902, 1909–13. *Source:* See Section II of The data and Appendix. *Note*: The German imports/UK imports are the total amount of German imports in millions of USD divided by the total amount of British imports in millions of USD. The German exports/UK exports are the total amount of German exports in millions of USD divided by the total amount of British exports in millions of USD

in the first four years of the 1880s, while in contrast British trade slowed down, even showing a reduction in growth in the same period (see table 1 and online appendix tables A4 and A7). By 1913, Germany had reduced the gap with Great Britain considerably, with British total exports and imports accounting for 7.136 million USD and German trade for 4.424 million USD (see online appendix tables A2 and A5).

Figures 2 and 3 illustrate the geographical distribution of German and British trade in the world in 1881 and 1913 (see also online appendix tables A3 and A6). The major part of German foreign commerce was directed to the European core countries, with Great Britain, Austria-Hungary, and France as its main trading partners. Likewise, British trade was concentrated in Europe but to a far lesser extent. In 1880, German and British trade with Europe accounted for more than 87 per cent and a little more than 41 per cent of total trade, respectively. At the time Germany intensified its external commercial relations, British foreign commerce was already more regionally diversified and present in international markets. Besides Europe, British trade concentrated on the prospering peripheral countries in the western and eastern hemispheres. In 1880, British trade with American countries accounted for 31 per cent and Asia for 15.5 per cent of the total British trade. They are followed by Australia and the African colonies, responsible for 5.05 per cent and 6.39 per cent of total trade, respectively. Yet, the global distribution of British foreign commerce remained relatively stable during the entire period of investigation. In 1913, British trade with Europe still accounted for about 31 per cent of total trade, followed by America with 27.64 per cent, and Asia with 16.48 per cent.

By the turn of the century, Germany began to expand its market shares, especially in South America and Asia, showing comparatively higher growth rates than British trade in these regions. As shown in table 1, between 1880 and 1913 German trade with South America increased by 9.76 per cent and trade with Asia increased by 9.11 per cent. British trade with South America and Asia, on the other hand, grew by 4.5 and 2.5, respectively. Interestingly, while German trade with

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TABLE 1 German and British trade by continent – compound annual average growth rates (%) – 1880–1900, 1900–1913, 1880–1913

1880–1900	**Germany**	**Great Britain**
Africa	11.23	3.69
Asia	10.05	0.22
Australia	15.62	0.36
Central America	15.34	−2.49
Europe	2.91	1.53
North America	7.33	1.18
Oceania	12.12	83.99
South America	11.37	2.81
Total	4.20	1.41
1900–1913		
Africa	10.92	5.54
Asia	8.38	6.39
Australia	6.61	3.93
Central America	5.06	5.79
Europe	3.87	3.26
North America	4.79	1.95
Oceania	109.35	5.05
South America	8.01	7.47
Total	4.92	4.03
1880–1913		
Africa	10.76	4.27
Asia	9.11	2.53
Australia	11.75	1.69
Central America	11.01	0.69
Europe	3.18	2.14
North America	6.15	1.43
Oceania	48.20	51.76
South America	9.76	4.47
Total	4.34	2.35

Source: See the data and appendix.

America constantly gained in importance and share relative to total trade, Great Britain eventually lost ground. German trade with America accounted for 9.75 per cent of total trade in 1880 and for 24.76 per cent in 1913. Trade with South American countries such as Argentina, Brazil, and Chile gained constantly in importance. Similarly, trade with Asia, in particular China, increased from 1.86 per cent of total trade in 1880 to 9.75 per cent in 1913. Trade with Europe diminished to a little bit more than 60 per cent of total trade in 1913.[47]

[47] The impressive growth rates of Oceania, mainly New Zealand and Australia, were the results of an increase from an initially nearly non-existent to a yet rather marginal trade between Germany and these countries (for growth rates, see

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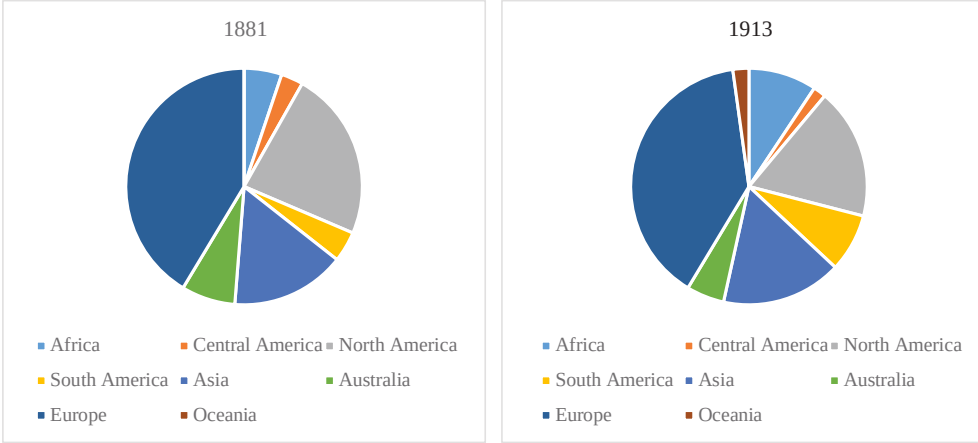

FIGURE 2 British trade by continent – share of total trade (%) – 1881 and 1913. *Sources*: See Section II of The data and Appendix [Colour figure can be viewed at wileyonlinelibrary.com]

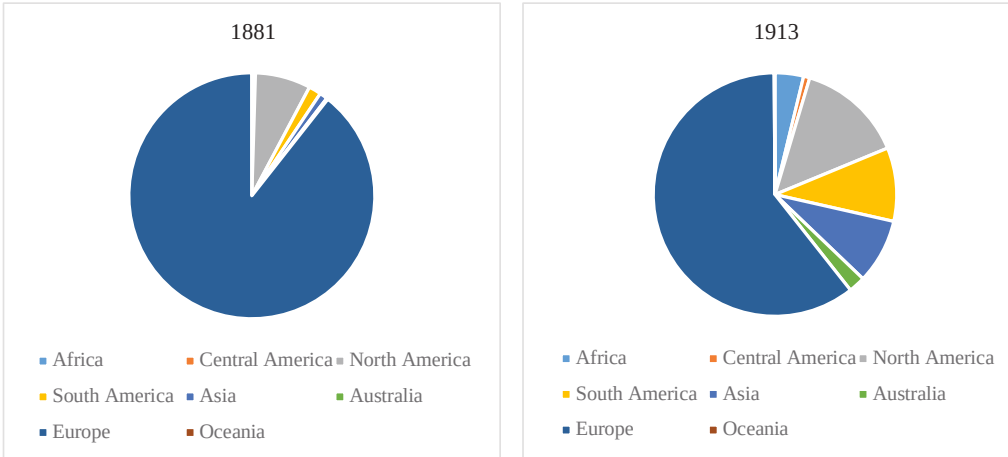

FIGURE 3 German trade by continent – share on total trade (%) – 1881 and 1913. *Sources*: See Section II of The data and Appendix [Colour figure can be viewed at wileyonlinelibrary.com]

The financial supremacy of Great Britain was equally striking. In 1881, a total number of 17 German banks with 19 branches were operating in 10 different countries. In the same year, 157 British banks with 400 branches were operating in 41 countries (see online appendix tables A8 and A9). Figures 4 and 5 show the geographical distribution of German and British banks in the world in 1881 and 1913. Similar to trade, British foreign banking was comparatively more diversified than German foreign banking, with a focus on the emerging economies of America and Asia, followed by Australia and the African colonies. Between 1881 and 1913, the percentage of banks operating in America in relation to the total number of banks increased slightly from 7 per cent to 9.4 per

also online appendix tables A4 and A7). In 1913, trade with Oceania accounted for 0.11% and trade with Australia for a little more than 2.2% of total German foreign trade.

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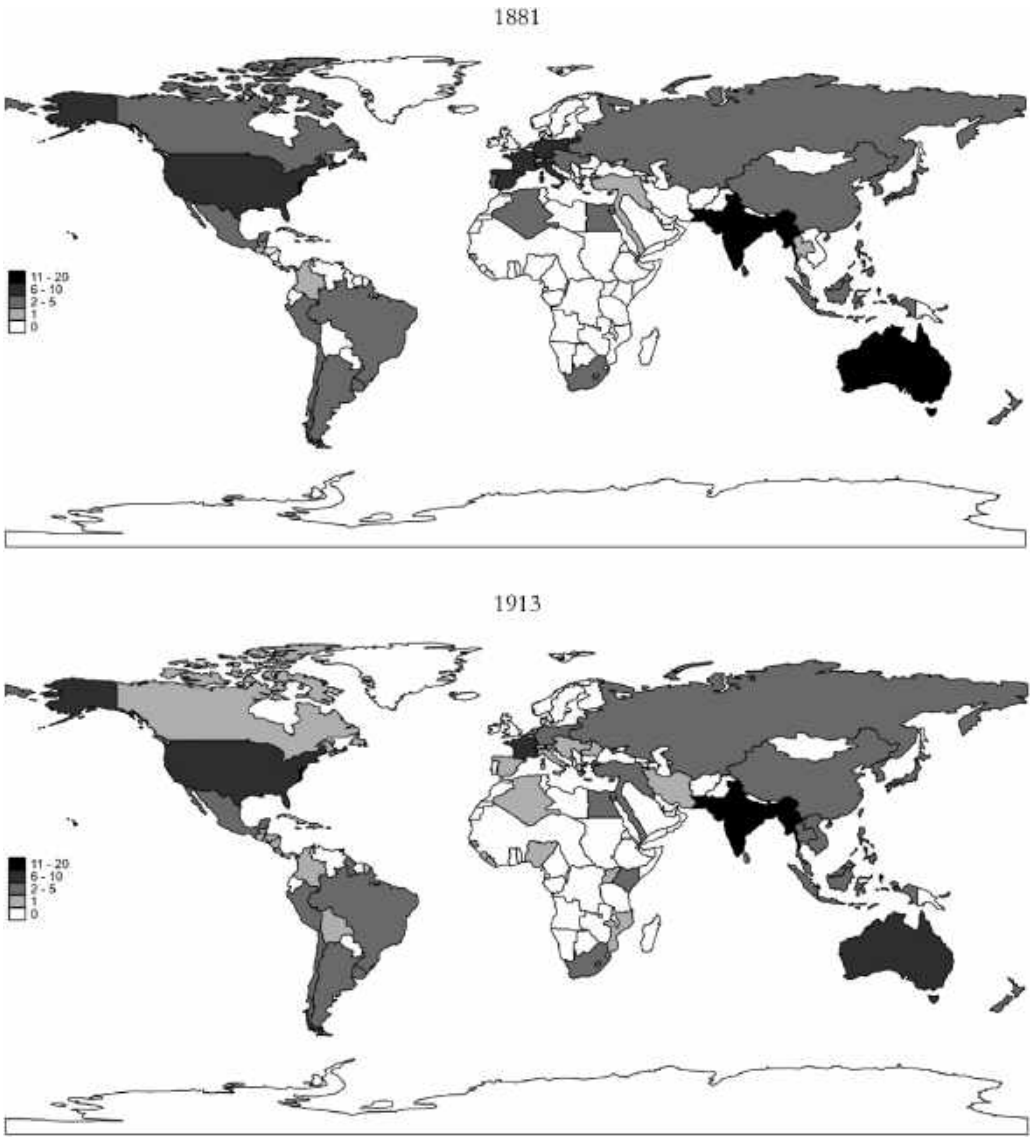

FIGURE 4 Geographical distribution of British foreign banks – number of banks by country – 1881 and 1913. *Sources*: See the data and appendix. *Note*: Owing to the scale of the map, the following British foreign bank presences are not depicted: in 1881, two banks in Malta, three banks in the British West Indies, and one in Mauritius, and in 1913, one bank in Antigua, Barbados, St Kitts, St Lucia, St Vincent, and the Grenadines; two banks in Mauritius; and four banks in Malta

cent and in Africa and Asia from 8.9 to 12.3 per cent and 29.9 to 34.8 per cent, respectively. In Australia, the share in total number of banks remained more or less the same, and in Europe, they even, although only marginally, decreased. Remarkably, British banks expanded their subsidiary network constantly and notably, especially in those countries of the Commonwealth that had adopted the 'Scottish' free banking model, such as Canada, New Zealand, and Australia (see online appendix tables A8–A11).

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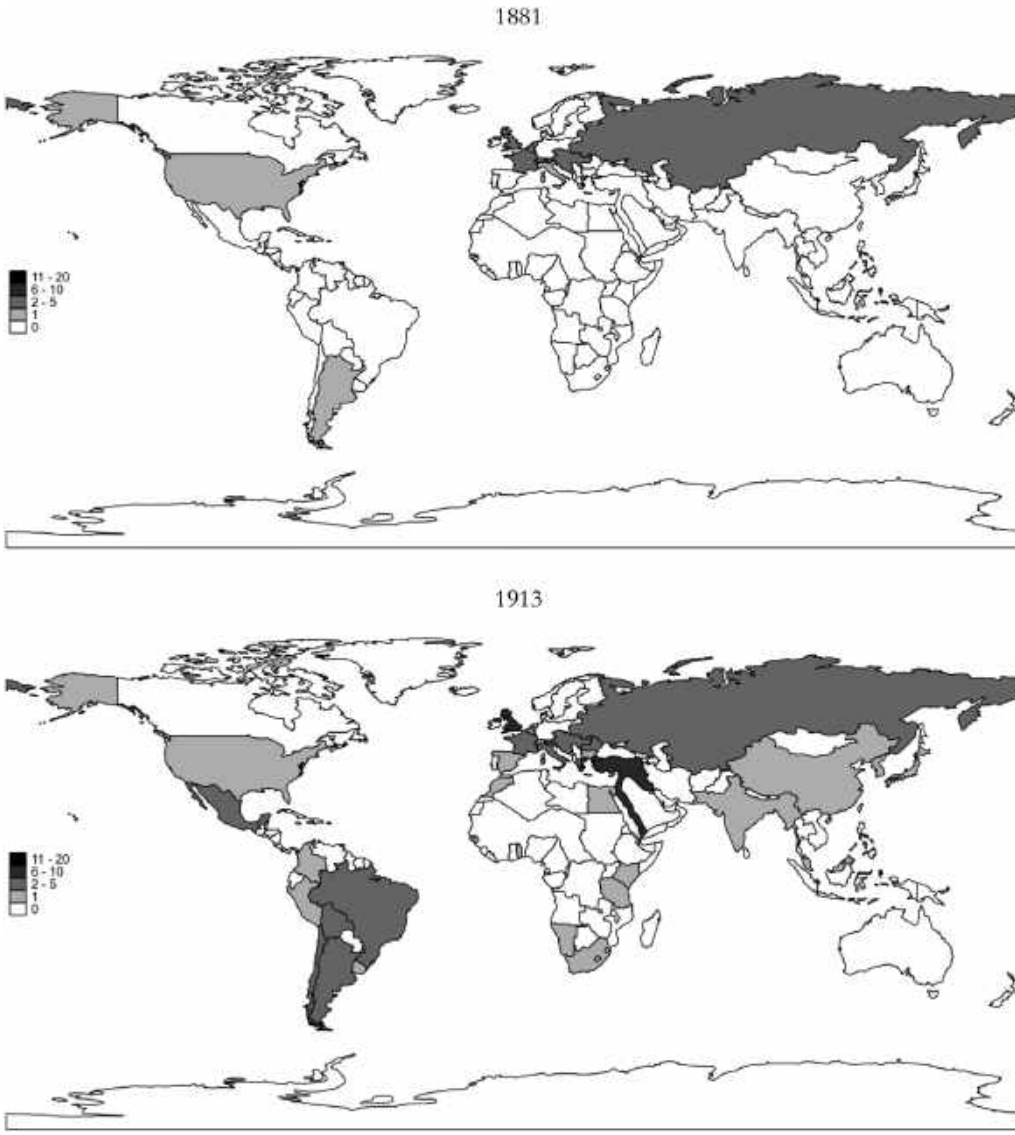

FIGURE 5 Geographical distribution of German foreign banks – number of banks by country – 1881 and 1913. *Sources*: See Section II of The data and Appendix

Yet, at the same time Germany managed to create its own network of overseas banking that stood in direct competition with the British banks. It seems that Germany was able to successfully penetrate the financial markets and establish a competitive banking network particularly in countries where its trade was only marginally developed [better, was initially limited], especially in Latin America. This reflects the idea of German banks being promoters of trade and contrasts with the British foreign banking dynamic. The primary purpose of the latter was to broaden already existing trade and engage in the increasing investment opportunities in the prospering peripheral markets and British colonies. Until the second half of the nineteenth century, British trade and its financing were managed by the merchant houses,

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maintaining offices in England and abroad. With British trade and investment accelerating intensively from the 1860s,[48] banks became indispensable to managing the increasing amount of financial transactions.[49] Following the premise '*Los von London*' and the aim to establish financial representation in the immediate neighbouring countries, the initial focus of German foreign banks was Great Britain and the other European core countries. However, over the 30-year period, there was a significant shift in the geographical distribution of the German banks, showing similar patterns to German foreign trade. While in 1881 more than 88 per cent of all German banks and their branches were located in Europe, by 1913 this figure had decreased to 45.2 per cent as the interest of German banks moved outside of Europe. The remaining 54.8 per cent were concentrated mainly – 73.5 per cent of the non-European share – in South and Central America and Asia.

In 1913, a total of 62 German banks were operating with 185 branches in 36 different countries. Still, Great Britain possessed more than twice as many foreign banks, 138, with 1523 branches in 63 countries. Yet, Germany caught up in establishing banks outside of Europe, in particular in the Latin American countries that, at the same time, were the focus of German foreign trade. Between 1862 and 1886 a total of 10 British banks were founded while Germany only established one (and that was liquidated in 1885). By the end of 1913, 17 German and 22 British Banks were operating in Latin America (see online appendix tables A8 and A9).

IV | BANKS AND TRADE – IS THERE A LINK?

This section investigates and compares the nexus of German and British world trade development and geographic banking penetration between 1881 and 1913 with an augmented trade gravity model.[50] It tests two principal hypotheses: (i) Germany and the United Kingdom traded relatively more with countries where German and British banks were present than with countries where they were not, and (ii) German banks initiated trade, and hence, the effect of German banks on trade is more pronounced in the years close to bank entry. Expressed in the form of equations, the model is as follows:

$$T_{jit} = \alpha_0 + \beta \mathrm{Ln}(Y_{it} + Y_{jt}) + \gamma \mathrm{Ln} D_{ij} + \delta B_{ijt} + \zeta BB_{ijt} + \eta \mathrm{Ln} S_{ijt} + \lambda_i + \lambda_j + \lambda_t + \varepsilon_{ijt} \quad (1)$$

$$T_{jit} = \alpha_0 + \beta \mathrm{Ln}(Y_{it} + Y_{jt}) + \gamma \mathrm{Ln} D_{ij} + \delta B_{ijt} + \zeta BB_{ijt} + \eta \mathrm{Ln} S_{ijt} + \boldsymbol{\psi}_{ijt} + \lambda_i + \lambda_j + \lambda_t + \varepsilon_{ijt} \quad (2)$$

[48] However, British banks had already expanded their activities to foreign countries in the 1820s. Founded in 1809 in London, the Nathan Mayer Rothschild & Sons Bank (NMR) started its operations in Brazil from independence in 1822. Initially concentrating on merchant banking activities, the NMR became the exclusive financial agent of the Brazilian government in London in 1855. They managed every Brazilian government fund and every purchase by the Brazilian government in England, as well as the payment of the dividends of Brazilian debt in London. See Marichal and Triner, 'European banks in Latin America', pp. 261–2.

[49] Besides being better suited to managing the increasing capital movements, British banks also released the merchant houses from the necessity of maintaining expensive representation abroad and thus enabled smaller firms to enter into the foreign trade business. Moreover, prevailing conditions in their home country triggered British banks to go abroad in the 1860s. In 1857–8, the British Parliament passed two company acts that extended the privilege of incorporation with limited liability to banks, leading to a boom of joint-stock bank foundations, many of them aiming for foreign markets. See Hurley, *Banking and credit*, p. 14; Young, 'British overseas banking', pp. 82–5; Briones and Villela, 'European bank penetration', p. 334.

[50] Head and Mayer, 'Gravity equations'.

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where T_{ijt} is the nominal value of bilateral trade (exports + imports) between country i and j in year t. α_0 is a constant. Y_{it} and Y_{jt} are the nominal national incomes of each country (GDP) in t, and D_{ij} is the great circle distance between the capital cities of i and j in kilometers (km). B_{ijt} and BB_{ijt} are numeric variables, representing the number of banks (B) and branches (BB) of country i present in country j at the time t, respectively. The variable S_{ijt} is the value of the average capital paid in of German and British banks in country j at the time t. It proxies the size of a bank. Two models are defined: (i) the 'baseline' model, Equation (1), that includes the explaining variables real GDP, distance, banks, bank branches, and bank size, and (ii) the 'extended' model, Equation (2), which differentiates between the overall effects of German versus British bank presence and the specific effect of German and British banks in each year. The variable ψ_{ijt} is a set of interactions, with the purpose of distinguishing between the effects of German and British banks, and their respective impact on trade in different years. Panel data are used to estimate Equations (1) and (2) with Poisson maximum likelihood (PPML).[51] The estimated models also include time (λ_t) and country fixed effects for each exporter and importer country (λ_i and λ_j), accounting for unobserved variables, such as cultural barriers and adherence to the gold standard, etc., that might influence the trade between two countries.[52] I also test the models (1) and (2) with different dependent variables: the nominal value of exports (X_{ijt}) and imports (X_{jit}), separately. For this study, the key coefficients of interest are δ and ζ, which show the impact of banks and their branches on trade.

The results for the bilateral trade estimation of the 'baseline' model, Equation (1), are presented in table 2 and for the 'extended' model, Equation (2), in table 3. The findings for the export and import estimations are presented accordingly in tables 4–5 and 6–7. They reflect the general theoretical assumptions for gravity models and confirm the main hypothesis of this study. While economic size has a positive impact on export volumes, higher distance leads to less trade. The key hypothesis of this paper, that banks matter, seems to be affirmed. The results, for each the 'baseline' model and the 'extended' models, show an overall significant and positive correlation between the number of banks and branches operating in a country and the respective German and British trade with that country.

With each additional bank in a country, both Germany and Great Britain's bilateral trade with that country increases by 12.7 per cent (table 2, column 1.1).[53] The effect of an additional branch is considerably smaller but still positively significant (table 2, column 1.2). The coefficient for the effect of banks on exports is larger than the coefficient for imports, suggesting that positive impact of banks on exports is larger than for imports, though only marginally (table 4, column 2.1 and table 6, column 3.1).

Bank size matters, too. The coefficient of bank size is positive and significant (table 2, columns 1.4–1.6), indicating that the larger the annual average paid-in capital of German and British banks operating in a country, the larger the bilateral trade between that country and Germany or Great Britain. Yet, these findings have to be taken with caution. Firstly, this is due to the limitations in the data (see section II). Secondly, the paid-in capital is a rather rough measure of bank size,

[51] Ibid.; Anderson and van Wincoop, 'Gravity with gravitas'.

[52] Santos Silva and Tenreyro, 'Log', have shown that ordinary least squares (OLS) estimates of log-linear models are likely inefficient, biased, and/or even inconsistent. They proposed PPML with robust standard errors as a superior alternative. Additionally, PPML allows for including '0' observations in the dependent variable, which have to be excluded or treated improperly under log-linear OLS estimates; see Lampe, 'Effects of bilaterialism'.

[53] See Wooldrige, *Introductory econometrics*.

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TABLE 2 Estimates of Equation (1) – German and British banks and trade – baseline model

	(1.1)	(1.2)	(1.3)	(1.4)	(1.5)	(1.6)
GDP	0.638***	0.795***	0.772***	0.766***	0.922***	0.872***
	(0.232)	(0.228)	(0.237)	(0.261)	(0.249)	(0.258)
Distance	−0.390	−0.660**	−0.454	−0.595	−0.835*	−0.634
	(0.310)	(0.314)	(0.294)	(0.467)	(0.447)	(0.434)
Banks	0.120***		0.0911***	0.0881***		0.0665*
	(0.0292)		(0.0300)	(0.0338)		(0.0342)
Branches		0.00592***	0.00398**		0.00472**	0.00352**
		(0.00217)	(0.00157)		(0.00209)	(0.00168)
Bank size				0.0412***	0.0439***	0.0341***
				(0.0128)	(0.00915)	(0.0121)
Year fixed effects	Y	Y	Y	Y	Y	Y
Country$_i$ fixed effects	Y	Y	Y	Y	Y	Y
Country$_j$ fixed effects	Y	Y	Y	Y	Y	Y
Observations	334	334	334	306	306	306

Source: Author's creation based on own calculations. See text.

Note: Robust standard errors are three-way clustered by country$_i$, country$_j$, and country pair, and are indicated in brackets. The variable *bank* is the number of British and German banks in country *j* in *t*. The variable *bank size* is the logarithmic value of the average capital paid in of German and British banks in country *j* in time *t*. The dependent variable is the sum of the value of exports from *i* to *j* in *t*, and the value of imports from *j* to *i* in *t* in millions of USD. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

not reflecting any dynamic changes in size owing to, for example, alteration of deposits or bank assets.[54]

The effect of banks on trade is more pronounced in the case of Germany. The positive significant coefficient of German banks on bilateral trade (table 3, column 1.7) is higher than the coefficient of the effect of their British counterparts (table 3, column 1.8). The same is true for exports and imports (table 5, column 2.8 and table 7, column 3.8). Most interestingly, German banks and trade seem to be more significantly and positively correlated in 1881 than in the following years. The coefficient for the interaction of the German bank with the year variable is larger in 1913 relative to previous years, suggesting that the impact of an additional German bank on trade in 1913 is relatively decreasing (1.15). The effect of British banks seems to be time invariant (table 3, column 1.16). The same patterns of contemporaneous bank effects apply to the influence of German and British banks on exports and imports (table 5, column 2.17 and table 7, column 3.17). This supports the second central idea of the different character and purpose of German and British banks. British banks mainly aimed to secure and maintain British hegemony in the international commercial markets. German banks, in contrast, were initiators of trade, with the primary task of supporting German commerce in gaining new markets and market shares abroad.

The finding that foreign banks have a significant positive influence on both the exports and imports from and to their homeland goes along with the common understanding of the literature on trade finance[55] and is supported by quantitative case evidence of German and British foreign

[54] Moreover, British banks kept significant amounts of unpaid capital as a form of contingent capital. See Turner, *Banking in crisis*.

[55] Kletzer and Bardhan, 'Credit markets'; Bordo and Rousseau, 'Historical evidence'.

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TABLE 3 Estimates of Equation (2) – German and British banks and trade – extended model

	(1.7)	(1.8)	(1.9)	(1.10)	(1.11)	(1.12)	(1.13)	(1.14)	(1.15)	(1.16)	(1.17)	(1.18)
GDP	0.538**	0.663***	0.511**	0.807***	0.491**	0.809***	0.568**	0.686***	0.788***	0.583*	0.708**	0.696**
	(0.219)	(0.228)	(0.223)	(0.231)	(0.223)	(0.235)	(0.229)	(0.234)	(0.294)	(0.324)	(0.358)	(0.280)
Distance	−0.587**	−0.440	−0.525	−0.676**	−0.544*	−0.516	−0.401	−0.456*	−0.384*	−0.399	−0.381*	−0.405*
	(0.266)	(0.348)	(0.385)	(0.316)	(0.295)	(0.323)	(0.257)	(0.264)	(0.226)	(0.256)	(0.227)	(0.246)
German banks	0.372***				0.329**		0.338***	0.284**		0.350***		
	(0.130)				(0.147)		(0.125)	(0.130)		(0.126)		
British banks		0.105***				0.0756**	0.0970***	0.0706**	0.101***			
		(0.0308)				(0.0321)	(0.0311)	(0.0324)	(0.0299)			
German branches			0.0388***		0.0163			0.00781				
			(0.0104)		(0.0123)			(0.0109)				
British branches				0.00574***		0.00426***		0.00387**				
				(0.00209)		(0.00152)		(0.00166)				
German banks × 1881									0.651***		0.694***	0.993***
									(0.141)		(0.136)	(0.301)
German banks × 1900									0.571***		0.555***	0.494***
									(0.161)		(0.158)	(0.182)
German banks × 1913									0.351***		0.345***	0.292***
									(0.1000)		(0.0988)	(0.104)
British banks × 1881										0.102***	0.120***	0.155***
										(0.0368)	(0.0372)	(0.0390)
British banks × 1900										0.0863***	0.0935***	0.0733**
										(0.0311)	(0.0299)	(0.0337)

(Continues)

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TABLE 3 (Continued)

	(1.7)	(1.8)	(1.9)	(1.10)	(1.11)	(1.12)	(1.13)	(1.14)	(1.15)	(1.16)	(1.17)	(1.18)
British banks × 1913										0.104***	0.0959***	0.0677**
										(0.0336)	(0.0337)	(0.0333)
German branches × 1881												−0.276*
												(0.152)
German branches × 1900												−0.000782
												(0.0476)
German branches × 1913												0.00949
												(0.0103)
British branches × 1881												−0.00772
												(0.00872)
British branches × 1900												0.00263
												(0.00257)
British branches × 1913												0.00312*
												(0.00173)
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country$_i$ fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country$_j$ fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	334	334	334	334	334	334	334	334	334	334	334	334

Source: Author's creation based on own calculations. See text.

Note: Robust standard errors are three-way clustered by country$_i$, country$_j$, and country pair, and are indicated in brackets. The variable *bank* is the number of British and German banks in country *j* in *t*. The variable *bank size* is the logarithmic value of the average capital paid in of German and British banks in country *j* in time *t*. Interactions between German and British banks and years are included. The dependent variable is the sum of the value of exports from *i* to *j* in *t*, and the value of imports from *j* to *i* in *t* in millions of USD. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

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TABLE 4 Estimates of Equation (1) – German and British banks and exports – baseline model

	(2.1)	(2.2)	(2.3)	(2.4)	(2.5)	(2.6)
GDP	0.591***	0.815***	0.779***	0.660***	0.868***	0.814***
	(0.193)	(0.216)	(0.217)	(0.205)	(0.208)	(0.210)
Distance	−0.199	−0.480	−0.282	−0.506	−0.735	−0.552
	(0.326)	(0.310)	(0.298)	(0.540)	(0.494)	(0.485)
Banks	0.130***		0.0942***	0.0939**		0.0664*
	(0.0341)		(0.0339)	(0.0390)		(0.0380)
Branches		0.00730***	0.00493***		0.00587***	0.00440***
		(0.00225)	(0.00141)		(0.00209)	(0.00144)
Bank size				0.0511***	0.0521***	0.0426***
				(0.0169)	(0.0128)	(0.0147)
Year fixed effects	Y	Y	Y	Y	Y	Y
Country$_i$ fixed effects	Y	Y	Y	Y	Y	Y
Country$_j$ fixed effects	Y	Y	Y	Y	Y	Y
Observations	334	334	334	306	306	306

Source: Author's creation based on own calculations. See text.

Note: Robust standard errors are three-way clustered by country$_i$, country$_j$, and country pair, and are indicated in brackets. The variable *bank* is the number of British and German banks in country *j* in *t*. The variable *bank size* is the logarithmic value of the average capital paid in of German and British banks in country *j* in time *t*. The dependent variable is the value of exports from *i* to *j* in *t* in millions of USD. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

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TABLE 5 Estimates of Equation (2) – German and British banks and exports – extended model

	(2.7)	(2.8)	(2.9)	(2.10)	(2.11)	(2.12)	(2.13)	(2.14)	(2.15)	(2.16)	(2.17)	(2.18)
GDP	0.484***	0.628***	0.491***	0.843***	0.466***	0.837***	0.532***	0.756***	0.776***	0.681**	0.803**	0.795***
	(0.170)	(0.192)	(0.172)	(0.225)	(0.171)	(0.222)	(0.185)	(0.208)	(0.232)	(0.293)	(0.325)	(0.267)
Distance	−0.382	−0.246	−0.366	−0.499	−0.369	−0.343	−0.201	−0.307	−0.181	−0.197	−0.179	−0.301
	(0.300)	(0.364)	(0.404)	(0.308)	(0.320)	(0.323)	(0.289)	(0.271)	(0.261)	(0.284)	(0.260)	(0.251)
German banks	0.324***				0.311**		0.287**	0.261**		0.308***		
	(0.125)				(0.141)		(0.119)	(0.117)		(0.117)		
British banks		0.120***				0.0811**	0.113***	0.0790**	0.117***			
		(0.0363)				(0.0368)	(0.0368)	(0.0369)	(0.0358)			
German branches			0.0260***		0.00530			−0.00448				
			(0.00963)		(0.0122)			(0.0109)				
British branches				0.00727***		0.00532***		0.00504***				
				(0.00227)		(0.00142)		(0.00149)				
German banks × 1881									0.597***		0.613***	0.933***
									(0.149)		(0.151)	(0.288)
German banks × 1900									0.467***		0.450***	0.482***
									(0.137)		(0.133)	(0.146)
German banks × 1913									0.295***		0.302***	0.272***
									(0.0906)		(0.0921)	(0.0937)
British banks × 1881										0.105***	0.123***	0.135***
										(0.0406)	(0.0411)	(0.0381)

(Continues)

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TABLE 5 (Continued)

	(2.7)	(2.8)	(2.9)	(2.10)	(2.11)	(2.12)	(2.13)	(2.14)	(2.15)	(2.16)	(2.17)	(2.18)
British banks × 1900										0.0992***	0.104***	0.0609
										(0.0379)	(0.0367)	(0.0378)
British banks × 1913										0.130***	0.125***	0.0811**
										(0.0368)	(0.0357)	(0.0356)
German branches × 1881												−0.334**
												(0.141)
German branches × 1900												−0.0849**
												(0.0379)
German branches × 1913												−0.00446
												(0.00998)
British branches × 1881												−0.00143
												(0.00957)
British branches × 1900												0.00519
												(0.00325)
British branches × 1913												0.00457**
												(0.00199)
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country$_i$ fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country$_j$ fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	334	334	334	334	334	334	334	334	334	334	334	334

Source: Author's creation based on own calculations. See text.

Note: Robust standard errors are three-way clustered by country$_i$, country$_j$, and country pair, and are indicated in brackets. The variable *bank* is the number of British and German banks in country *j* in *t*. The variable *bank size* is the logarithmic value of the average capital paid in of German and British banks in country *j* in time *t*. Interactions between German and British banks and years are included. The dependent variable is the value of exports from *i* to *j* in *t* in millions of USD. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

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TABLE 6 Estimates of Equation (1) – German and British banks and imports – baseline model

	(3.1)	(3.2)	(3.3)	(3.4)	(3.5)	(3.6)
GDP	0.696**	0.814***	0.796***	0.861**	0.985***	0.935***
	(0.307)	(0.293)	(0.306)	(0.348)	(0.333)	(0.344)
Distance	−0.564*	−0.821**	−0.616*	−0.662	−0.898**	−0.696
	(0.324)	(0.337)	(0.317)	(0.463)	(0.448)	(0.444)
Banks	0.107***		0.0839***	0.0802***		0.0630**
	(0.0263)		(0.0277)	(0.0303)		(0.0318)
Branches		0.00485**	0.00328*		0.00382*	0.00287
		(0.00221)	(0.00176)		(0.00220)	(0.00190)
Bank size				0.0351***	0.0388***	0.0291**
				(0.0121)	(0.00952)	(0.0122)
Year fixed effects	Y	Y	Y	Y	Y	Y
Country$_i$ fixed effects	Y	Y	Y	Y	Y	Y
Country$_j$ fixed effects	Y	Y	Y	Y	Y	Y
Observations	334	334	334	306	306	306

Source: Author's creation based on own calculations. See text.

Note: Robust standard errors are three-way clustered by country$_i$, country$_j$, and country pair, and are indicated in brackets. The variable *bank* is the number of British and German banks in country *j* in *t*. The variable *bank size* is the logarithmic value of the average capital paid in of German and British banks in country *j* in time *t*. The dependent variable is the value of imports from *j* to *i* in *t* in millions of USD. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

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TABLE 7 Estimates of Equation (2) – German and British banks and imports – extended model

	(3.7)	(3.8)	(3.9)	(3.10)	(3.11)	(3.12)	(3.13)	(3.14)	(3.15)	(3.16)	(3.17)	(3.18)
GDP	0.596*	0.715**	0.551*	0.819***	0.529*	0.822***	0.614*	0.663**	0.830**	0.545	0.691*	0.679**
	(0.306)	(0.300)	(0.308)	(0.293)	(0.315)	(0.300)	(0.314)	(0.320)	(0.384)	(0.370)	(0.396)	(0.335)
Distance	−0.768***	−0.619*	−0.655*	−0.835**	−0.691**	−0.681**	−0.590**	−0.599**	−0.575**	−0.590**	−0.574**	−0.518**
	(0.258)	(0.357)	(0.382)	(0.341)	(0.292)	(0.344)	(0.256)	(0.278)	(0.224)	(0.257)	(0.226)	(0.259)
German banks	0.419***				0.344**		0.388***	0.304**		0.392***		
	(0.138)				(0.156)		(0.134)	(0.145)		(0.137)		
British banks		0.0897***				0.0659**	0.0801***	0.0575*	0.0842***			
		(0.0276)				(0.0294)	(0.0275)	(0.0294)	(0.0264)			
German branches			0.0506***		0.0262*			0.0191				
			(0.0129)		(0.0141)			(0.0129)				
British branches				0.00459**		0.00350**		0.00302				
				(0.00204)		(0.00167)		(0.00186)				
German banks × 1881									0.710***		0.784***	1.064***
									(0.148)		(0.133)	(0.323)
German banks × 1900									0.692***		0.681***	0.530**
									(0.188)		(0.189)	(0.225)
German banks × 1913									0.408***		0.388***	0.304***
									(0.113)		(0.109)	(0.116)
British banks × 1881										0.0952***	0.114***	0.161***
										(0.0350)	(0.0357)	(0.0413)
British banks × 1900										0.0711***	0.0815***	0.0730**
										(0.0268)	(0.0256)	(0.0317)

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TABLE 7 (Continued)

	(3.7)	(3.8)	(3.9)	(3.10)	(3.11)	(3.12)	(3.13)	(3.14)	(3.15)	(3.16)	(3.17)	(3.18)
British banks × 1913										0.0770**	0.0639*	0.0455
										(0.0334)	(0.0356)	(0.0335)
German branches × 1881												−0.229
												(0.168)
German branches × 1900												0.0708
												(0.0651)
German branches × 1913												0.0228*
												(0.0119)
British branches × 1881												−0.0117
												(0.00862)
British branches × 1900												0.00107
												(0.00222)
British branches × 1913												0.00214
												(0.00171)
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country$_i$ fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country$_j$ fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	334	334	334	334	334	334	334	334	334	334	334	334

Source: Author's creation based on own calculations. See text.

Note: Robust standard errors are three-way clustered by country$_i$, country$_j$, and country pair, and are indicated in brackets. The variable *bank* is the number of British and German banks in country *j* in *t*. The variable *bank size* is the logarithmic value of the average capital paid in of German and British banks in country *j* in time *t*. Interactions between German and British banks and years are included. The dependent variable is the value of imports from *j* to *i* in *t* in millions of USD. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

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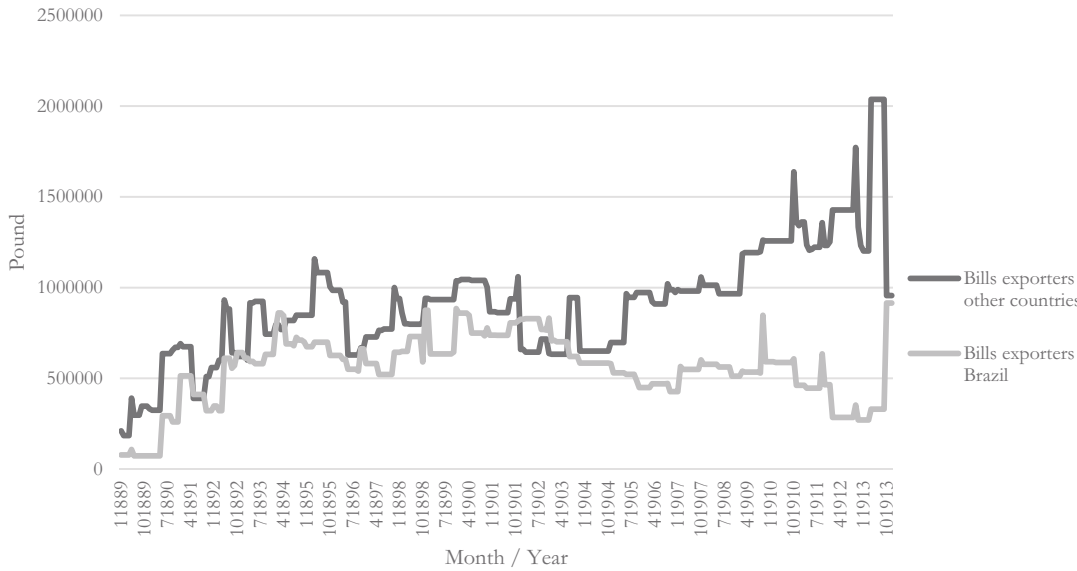

FIGURE 6 Authorized credit lines - bills discounted - by *Brasilianische* for exporters from Brazil and exporters from other countries – Monthly value in Pound – 1889–1913. *Sources*: See Appendix

banks in Latin America. The two nations established the most dominant and influential foreign financial institutions of the late nineteenth and early twentieth centuries on the continent, focusing on the three largest and most developed economies, Argentina, Brazil, and Chile.[56] The Brasilianische Bank für Deutschland, founded in 1887, was the leading German bank in Brazil. An analysis of its annual banking statements between 1887 and 1913 confirms that the bank financed both companies exporting from Brazil to other countries and companies exporting from other countries to Brazil. Figure 6 shows the monthly amount of bills of exchange from exporting companies in Brazil and exporters from other countries discounted by the Brasilianische Bank für Deutschland.

Yet, this study suffers from the traditional disadvantage of a possible inconsistent parameter estimation owing to endogenous regressors. In other words, although disclosing clear tendencies, it is limited in its capacity to provide axiomatic empirical information on what comes first, banks or trade. An attempt has been made to remedy this problem by including time effects, as well as importer and exporter fixed effects. The best option from a theoretical and empirical perspective would be the inclusion of an instrumental variable, accounting for the establishment of banks not driven by trade developments. The identification and definition of such an instrument remains an unsolved challenge. However, there are some indicators that provide qualitative evidence on the reasons why German banks entered a certain country, establishing a certain causal chain between banking internationalisation and trade development.

In the founding memorandum of the German East Africa Bank (Deutsch-Ostafrikanische Bank), a German banker highlighted the negative effect the absence of German banks abroad and the resulting German dependency on foreign finance had on German commerce. 'The lack of a (German) bank in German East Africa has been felt as a disadvantage for a long time. […]

[56] Kisling, 'La financiación del comercio'.

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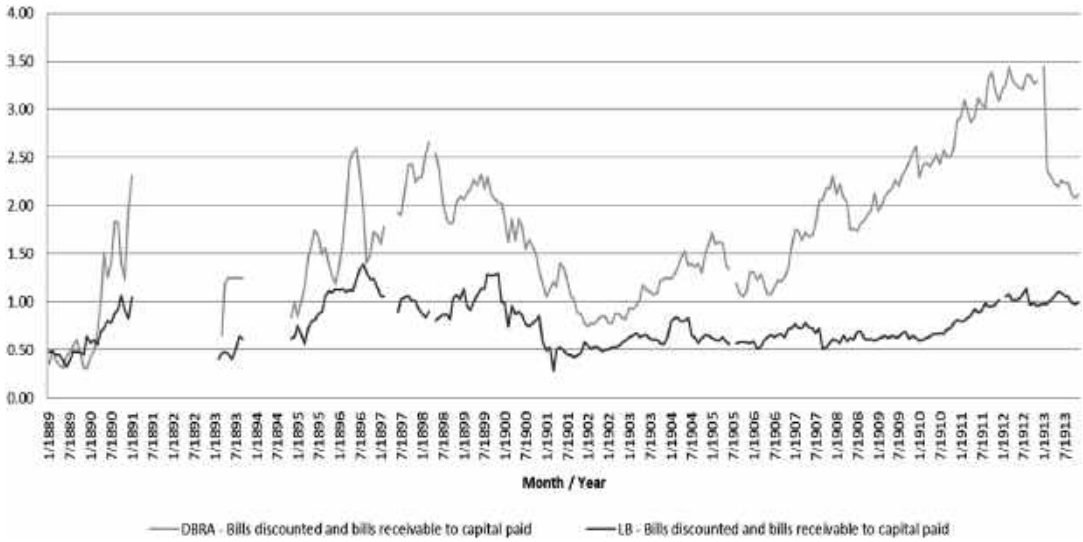

FIGURE 7 Bills discounted and bills receivable to capital paid in – Brasilianische Bank für Deutschland (DBRA) and London & Brazilian Bank de Rio de Janeiro (LB) – monthly relationship, 1889–1913. *Source*: Kisling, 'La financión del comercio', p. 198

it contributed significantly to increasing the commercial dependence of the German protectorate on Zanzibar [...] that disposes over branches of some British Indian banks (Chartered Bank of India, National Bank of India).'[57]

The negative effect of German bank absence on trade becomes even more evident when considering that German banks tended to provide longer credit terms than their British, French, or US competitors, and they furthermore directed more of their resources towards trade finance than banks of other nationalities.[58] Figure 7 compares the ratio of trade finance to capital of German and British banks in Brazil between the entry of German banks in 1889 and 1913.

Similarly, the Brasilianische Bank für Deutschland stressed in its memorandum that one of the bank's priorities was to expand its business abroad in order to '[...] be able to serve the motherland as precious pioneer [...]'.[59] This central idea of German banks paving the way for Germany's commercial expansion in overseas markets has indeed been subject to several contemporaneous studies on German foreign banking.

> 'We already pointed out that foreign banks are first and foremost concerned with the prosperity of their national customers. The German banks have progressed along this path and in the course of time have assumed the role of true 'pioneers' of the German economy. [...] They had first established a generous overseas information and news service and in this way aroused the interest of their customers at home for the still unconquered economic areas, and then, as a next step, sought to

[57] Hauser, 'Die Deutschen Überseebanken', p. 73, translated from German by the author.

[58] Wolfe, 'Foreign credits', p. 18; Kisling, 'La financiación del comercio', p. 198.

[59] Ahrens, *Die deutschen Auslandsbanken,* p. 38, translated from German by the author.

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systematically draw the attention of foreign customers in the overseas countries (these are the ones we are mainly dealing with here) to German houses. In individual cases, it is said that the foreign banks even went so far as to become regular agencies of German export and industrial companies. In this way, many a customer has been won for the German houses and old German sales areas have been significantly expanded.'[60]

V | CONCLUDING REMARKS

The aim of this study has been to investigate the role of German and British foreign banks in the development of German and British foreign trade in the late nineteenth and early twentieth centuries, following the principal hypothesis that banks promoted international trade by providing financial and informational assistance abroad.

A novel data set was created on German and British foreign banks and world trade in the years between 1881 and 1913. Using an augmented gravity model of trade, the study investigated the relation between the geographical banking distribution and trade developments of the two countries. The results confirm a significant and positive correlation between banks and trade; Germany and Great Britain traded more with countries where banks were present. Most interestingly, German banks and trade seem to be more significantly and more highly correlated in the 1880s, the years of the establishment of the first German foreign banks, than in the following years. In contrast, the effect of British banks seems to be constant over time.

This study contributes to the literature on trade finance, providing quantitative evidence on a positive relationship between foreign bank penetration and export developments. Moreover, the findings not only lend empirical support to the literature on German foreign banking during the period of the first globalisation but also to the study of the role of finance in the development of the foreign trade of a 'latecomer' in comparison with an economy already established in the international markets. Qualitative studies affirm that the primary purpose of the *Auslandsbanken* was to provide financial assistance to trade in those markets where German foreign commerce was underrepresented. They hence acted as promoters of trade. In contrast, the primary purpose of British foreign banks was to broaden already existing trade and engage in increasing investment opportunities in the prospering peripheral markets and British colonies. Thus, the effect of bank entry on trade was initially more significant in case of Germany.

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[60] Strasser, *Die Deutschen Banken im Ausland*, pp. 24–5, translated from German by the author.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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APPENDIX A: SOURCES

The data on monthly credits provided by Brasilianische Bank für Deutschland between 1888 and 1913 have been compiled from the official reports of the advisory board meetings (*Aufsichtsratberichte*) from 1888 to 1913.[61]

Real GDP data was obtained from the Maddison Project Database, Version 2013.[62] In the cases of South and Central America, Africa, and Asia, the data provided do not cover the entire years of observation (1881, 1900, and 1913), and missing years are estimated by geometric interpolation. The real GDP of Austria-Hungary is estimated as a weighted sum of GDP of Austria and Hungary. The GDP for West African Settlements is the GDP of Ghana.

The calculations on great circle distance in kilometers are based on the data on the geographic coordinates of the countries capitals (longitude and latitude) provided by the Global Gazetteer, version 2.3.[63] Exchange rates were extracted from Denzel.[64]

The classification of continents is as follows: (i) Europe includes the countries Austria-Hungary, Belgium, Bulgaria, Cyprus, Denmark, Finland, Germany, Great Britain, Greece, Italy, the Netherlands, Norway, Portugal, Romania, Russia (including Poland), Spain, Sweden, Switzerland, Turkey (Ottoman Empire), and Yugoslavia (Montenegro and Serbia). (ii) Africa includes the countries Algeria, Angola, Burundi, Cameroon, Congo, Egypt, Equatorial Guinea, Ethiopia, German Southwest Africa, Gibraltar, Guinea, Kenya (British East Africa), Liberia, Madagascar, Malta, Mauritius, Morocco, Mozambique, Niger, the Protectorate of Northern Nigeria, the Protectorate of Southern Nigeria, Senegal, South Africa, Tanzania (German East Africa), Togo, Uganda, and West African Settlements. (iii) Asia includes the countries China, Hong Kong, India, Indonesia, Persia (Iran), Japan, Jemen (Aden), Malaysia (British East Indies), Myanmar (Burma), Palestine, Philippines, Singapore (the Strait Settlements), Sri Lanka (Ceylon), Syria, Thailand (Siam), Tunisia, and Vietnam (Indochina). (iv) Central America includes the countries British West Indies (without St Lucia, St Kitts and Nevis, St Vincent and the Grenadines, Trinidad and Tobago, and the Virgin Islands), Cuba, El Salvador, Haiti, Honduras, Mexico, Nicaragua, and Panama. (v) North America

[61] Brasilianische Bank für Deutschland, 'Sitzung des Aufsichtsrats – Sitzungssaal Norddeutsche Bank Hamburg' (1888–1913), KA/799, Historische Archiv der Deutschen Bank, Frankfurt am Main.

[62] Maddison Project Database, version 2013. Bolt and van Zanden, *First update*.

[63] Global Gazetteer, ver. 2.3, http://www.fallingrain.com/world/index.html (accessed 6 Sept. 2022).

[64] Denzel, *Handbook*.

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includes the countries Antigua, the Bahamas, Barbados, Belize, Bermuda, Canada, Costa Rica, Dominican Republic, Jamaica, and the United States of America (without the state of Hawaii), (vi) South America includes the countries Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Guatemala, Guyana (British Guiana), Paraguay, Peru, Uruguay, and Venezuela. (vii) Australia is Australia. (viii) Oceania includes the US state Hawaii and the countries New Zealand, Papa New Guinea, St Lucia, St Kitts and Nevis, St Vincent and the Grenadines, Trinidad and Tobago, and the Virgin Islands.

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